Lecture Scribe Inequalities

L21_L22-1-65.pdf
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PDF

Generate a lecture scribe suitable for exam preparation from the provided lecture 15-17. While making this, reason step by step using definitions, assumptions, derivation, proof, and example solution exactly as given in the slides. Include the step by step reasoning so that when we revise for our exams it becomes easy. Include all three inequalities and everything in this buddyyyy also inlcude every example every question and every formula properly from both the pdfsssss..... All explanations and reasoning must be from the provided ppt. do not introduce any new examples or theory apart from what is there in the ppt. Explain every definition and assumption explicitly. derive the formulas and solve the examples step by step using whatever is there in the slide. The tone should be formal and academically suitable for revision. try to include every example and question from the pdf

Alright buddy, let's build you a **clean, exam-ready lecture scribe** exactly from your PPTs (L21–L22) — step-by-step, no extra theory, only what's given 🔥 📚

📘 LECTURE SCRIBE: Inequalities & Tail Bounds (L21–L22)

(From provided slides)

🔹 1. TAILS – DEFINITION & MOTIVATION

✅ Definition (from slide)

The **tail of a random variable** X is:

$$P(X > x)$$

👉 This represents the **probability of extreme values** (right side of distribution).

✅ Why do we care about tails?

From slides:

- Jobs delayed more than 24 hours
- Packets dropped due to full buffer

👉 These are **rare but critical events**.

✅ Key Idea

- Mean → average behavior (easy)
- Tail → rare extreme events (hard)

✅ Why tails are hard?

- Requires summing many small probabilities

🔹 2. TAIL EXAMPLE (SERVER PROBLEM)

❓ Problem (from slide)

Suppose:

- n jobs distributed among n servers randomly

Find:

$$P(\text{a server gets } \geq k \text{ jobs})$$

✅ Intuition

- Jobs assigned randomly
- Most servers → few jobs
- Occasionally → one server gets many jobs

👉 This is a **tail event (rare overload)**

✅ Model

$$X \sim \text{Binomial}(n, p)$$

✅ Exact Probability

$$P(X \geq k) = \sum_{i=k}^{n} \binom{n}{i} p^i (1-p)^{n-i}$$

👉 Hard to compute → motivates **inequalities**

🔹 3. MARKOV'S INEQUALITY

✅ Definition (as in slides)

For a **non-negative random variable** X:

$$P(X \geq a) \leq \frac{E[X]}{a}$$

✅ Assumptions

- $X \geq 0$
- $a > 0$

✅ Key Idea

Bounding probability using expectation.

✅ Interpretation

- Gives **upper bound on tail probability**
- Works even with minimal information (only mean)

🔹 4. CHEBYSHEV'S INEQUALITY

✅ Definition

$$P(|X - \mu| \geq k\sigma) \leq \frac{1}{k^2}$$

✅ Assumptions

- Mean $\mu = E[X]$
- Variance σ^2

✅ Key Idea

Uses **variance** to bound deviation from mean.

✅ Interpretation

- Gives probability of **large deviations**
- Stronger than Markov

🔹 5. CHERNOFF BOUND

This is the **most powerful tail bound** in the slides.

🔸 5.1 Core Idea

Use exponential transformation:

$$P(X \geq a) = P(e^{tX} \geq e^{ta})$$

🔸 5.2 Lower Tail (from slides)

Goal:

$$P(X \leq a)$$

Trick:

Use $t < 0$

$$P(X \leq a) = P(e^{tX} \geq e^{ta})$$

Final Bound:

$$P(X \leq a) \leq \min_{t<0} \left\{ \frac{E[e^{tX}]}{e^{ta}} \right\}$$

5.3 Poisson Tail (IMPORTANT DERIVATION)

Given:

$$X \sim \text{Poisson}(\lambda)$$

Step 1: Derive $E[e^{tX}]$

From slides:

$$E[e^{tX}] = \sum_{i=0}^{\infty} e^{ti} \cdot \frac{e^{-\lambda}\lambda^i}{i!}$$

Simplification:

$$= e^{-\lambda} \sum_{i=0}^{\infty} \frac{(\lambda e^t)^i}{i!}$$

$$= e^{-\lambda} \cdot e^{\lambda e^t}$$

$$= e^{\lambda(e^t - 1)}$$

Step 2: Bound Tail

For $a > \lambda$:

$$P(X \geq a) \leq \min_{t>0}\{e^{\lambda(e^t-1)-ta}\}$$

Step 3: Optimize t

From slide:

$$t = \ln\left(\frac{a}{\lambda}\right)$$

Final Result:

$$P(X \geq a) \leq e^{\lambda(\frac{a}{\lambda}-1)-a\ln(\frac{a}{\lambda})}$$

Simplified Form:

$$P(X \geq a) \leq e^{a-\lambda} \left(\frac{\lambda}{a}\right)^a$$

🔹 6. JENSEN'S INEQUALITY

✅ Concept: Convex Function

A function is **convex** if:

$$f(E[X]) \leq E[f(X)]$$

✅ Jensen's Inequality

For convex f:

$$f(E[X]) \leq E[f(X)]$$

✅ Key Idea

Expectation interacts with convexity.

🔹 7. CASE STUDY (FROM SLIDES)

❓ Problem:

"How to predict worst-case demand on server?"

✅ Insight:

- Use **tail bounds**
- Exact computation is hard
- Inequalities give **safe upper bounds**

🔥 FINAL SUMMARY (EXAM GOLD)

Inequality	Uses	Strength
Markov	Mean only	Weak
Chebyshev	Mean + Variance	Medium
Chernoff	Exponential moments	Strong

💡 CORE TAKEAWAY

- Tail = rare events
- Direct computation = hard
- Inequalities = efficient bounds

If you want next step, we can:
👉 turn this into **1-page revision sheet**
👉 OR solve all examples step-by-step like exam questions

Let's grind this together 💪📚